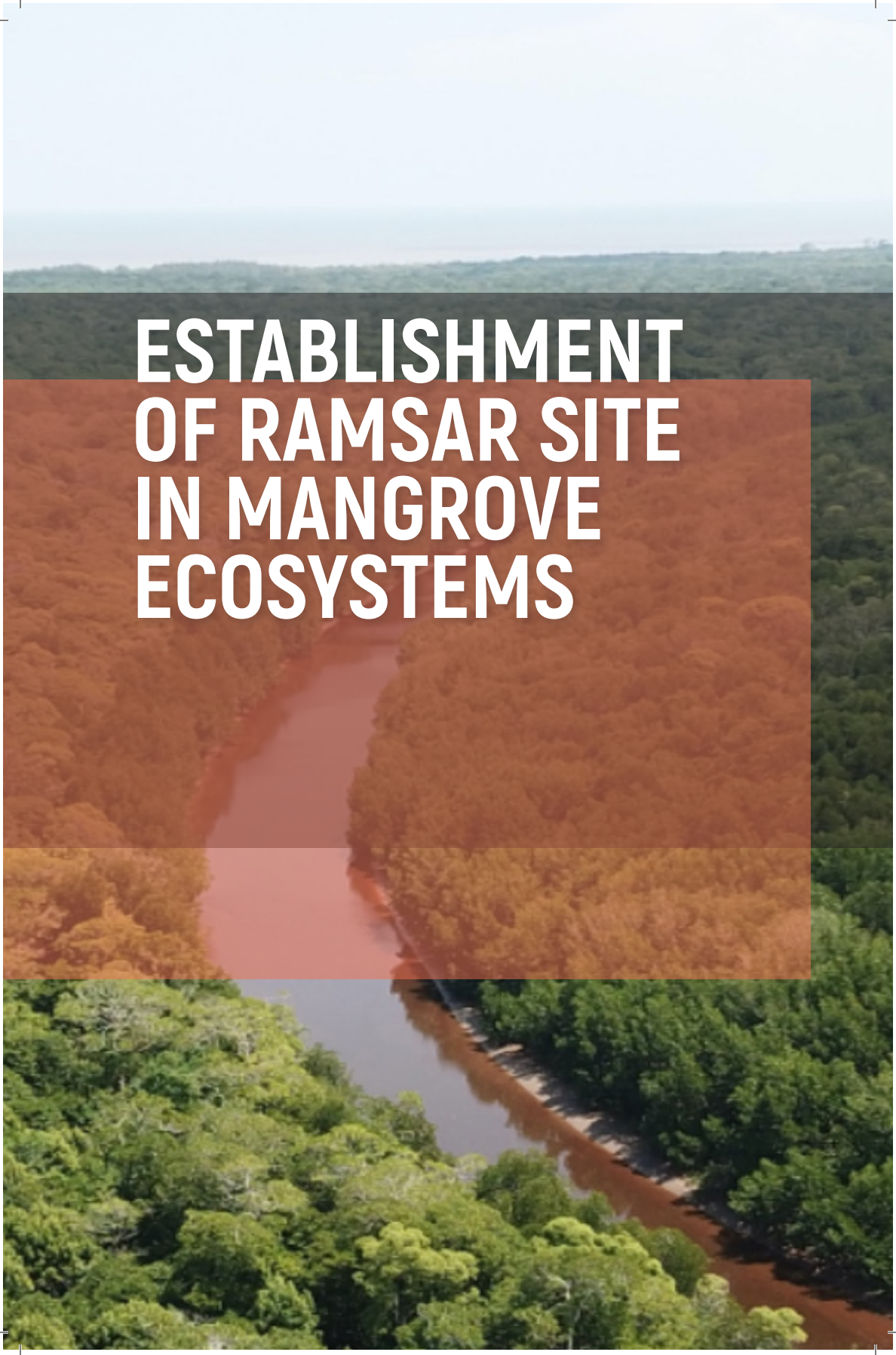


ESTABLISHMENT OF RAMSAR SITE IN MANGROVE ECOSYSTEMS

ESTABLISHMENT OF RAMSAR SITE IN MANGROVE ECOSYSTEMS

Ahmad Faisal Siregar
Fairus Mulia
Nyoto Santoso

Penerbit IPB Press
Taman Kencana Street No. 3,
Bogor City - Indonesia

C.01/12.2025

Book Title:
ESTABLISHMENT OF RAMSAR SITE IN MANGROVE ECOSYSTEMS

Authors:
Ahmad Faisal Siregar
Fairus Mulia
Nyoto Santoso

Editors:
Nyoto Santoso
Ahmad Faisal Siregar

Proof Reader:
Atika Mayang Sari

Cover Design & Layouter:
Rika Novida
Raisya Andarwina Ilmar
Alfyandi

Number of Pages:
32 + 10 roman pages

Edition/Printed:
1st Printed, December 2025

Citation:
Siregar, A. F., Mulia, F., & Santoso, N. (2025). Establishment of RAMSAR Site in Mangrove Ecosystems. Guideline for ASEAN Strategy on Mangrove Ecosystem Management, Series 10. ASEAN Mangrove Network (AMNET).

Published and Printed by:
PT Penerbit IPB Press
Member of IKAPI
Taman Kencana Street No. 3, Bogor 16128
Phone. 0251 - 8355 158 E-mail: ipbpress@apps.ipb.ac.id
www.ipbpress.com

© 2025, COPYRIGHT PROTECTED BY THE LAW
Do not cite or copy part or the entire book without written permission from the publisher

TABLE OF CONTENTS

TABLE OF CONTENTS v

FOREWORD vii

I. INTRODUCTION 1

A. Background 1

B. Goal and Objectives 3

C. Scope 4

D. Glossary 4

II. RAMSAR CONVENTION 7

A. Background of RAMSAR Site 7

B. Definition of RAMSAR Site 9

C. Basis For Determining RAMSAR Sites 10

D. Purpose of Determining A RAMSAR Site 12

E. Types and Classification of Wetlands In RAMSAR 12

F. Area and RAMSAR Site 16

G. Potential of Mangrove as A RAMSAR Site 17

III. DETERMINATION OF RAMSAR SITES 21

A. Determination Criteria 21

B. RAMSAR Documents 24

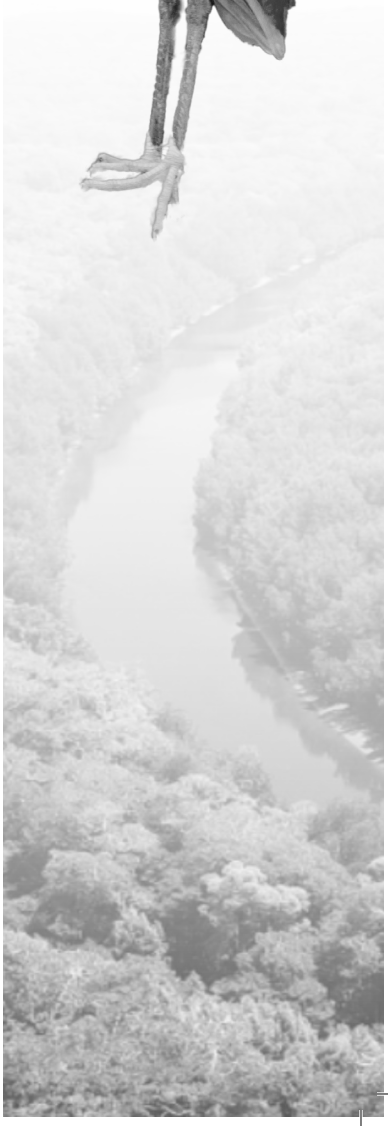
C. Management Plans 26

IV. CLOSING 29

REFERENCES 31

FOREWORD

Mangrove ecosystems are an integral part of the ASEAN region's coastal landscapes, delivering vital ecological, economic, and social benefits. They provide critical habitats for biodiversity, protect shorelines from erosion, buffer communities from storms, sustain coastal fisheries and livelihoods, and serve as one of the world's most important blue carbon ecosystems alongside seagrass beds and salt marshes. With Southeast Asia hosting the largest extent of mangrove globally, their conservation and sustainable management are of regional and global significance. Yet, these ecosystems continue to face pressures from unsustainable exploitation, land conversion, climate change impacts, and weak governance, underscoring the urgent need for coordinated, science-based, and inclusive management approaches across ASEAN.



The designation of Ramsar Sites is a key international mechanism for protecting wetlands of outstanding ecological value, including mangrove ecosystems. Established under the Ramsar Convention, the designation promotes the conservation and wise use of wetlands through national action and international cooperation. Ramsar Sites are recognized for conserving biodiversity, particularly migratory and waterbird habitats, while also contributing to water regulation, shoreline stabilization, and carbon storage. By elevating their ecological and socio-economic importance, Ramsar designation strengthens their role in national planning and global environmental commitments.

In the ASEAN context, the establishment of Ramsar Sites within mangrove ecosystems directly supports Strategy No. 3: Strengthening governance as well as establishing and implementing appropriate policy related to mangrove ecosystem management in each AMS. Ramsar designation reinforces legal and institutional frameworks, provides a mechanism for policy integration, and enhances accountability in wetland management. It also strengthens coordination among agencies, communities, and international partners while aligning national commitments with global agendas such as the Sustainable Development Goals, the Convention on Biological Diversity, and the United Nations Decade on Ecosystem Restoration.

This technical guideline has been developed to support ASEAN Member States by offering practical direction for the effective management of existing Ramsar Sites and guidance for identifying, proposing, and designating new ones. It consolidates regional experiences, best practices, and practical tools to assist policymakers and stakeholders in ensuring that Ramsar Sites contribute to biodiversity conservation and sustainable

development. Through this guideline, ASEAN reaffirms its collective commitment to safeguarding mangrove ecosystems as vital natural capital, ensuring that their benefits endure for generations to come

Author

Indonesia, 2025

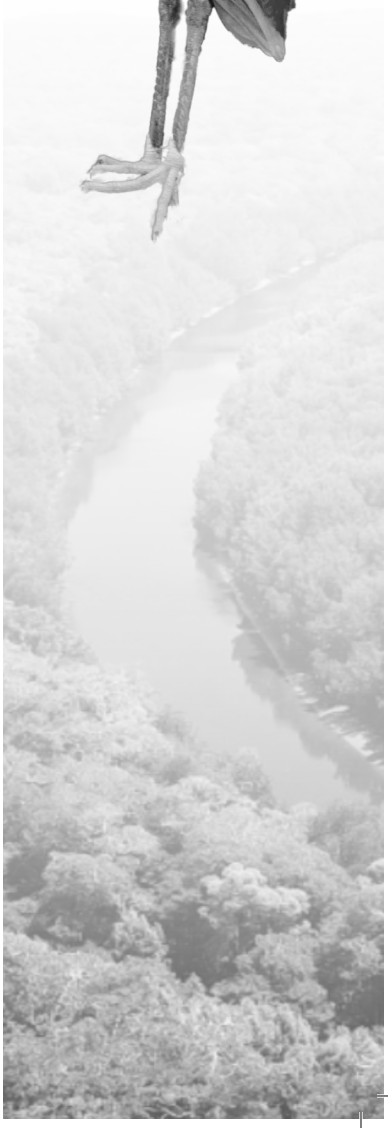
I.

INTRODUCTION

A. Background

Mangroves are unique coastal vegetation characterized by specialized rooting systems that enable them to adapt to tidal areas with muddy or sandy substrates (Macnae, 1968; Snedaker, 1978; Aksornkoae, 1993). The comprehensive scope of mangrove resources generally encompasses the mangrove ecosystem, which includes:

1. One or more species of trees and shrubs whose life is exclusively limited to mangrove habitats (exclusive mangroves).
2. Plant species that thrive in mangrove habitats but can also exist in non-mangrove environments (non-exclusive mangroves);
3. Biota is associated with mangroves (terrestrial and marine organisms, crustacean mosses, fungi, algae,



bacteria, etc.) that are permanent, temporary, occasional, or regularly found, and those specifically living in mangrove habitats.

4. Processes that sustain the ecosystem within and beyond the vegetation area.
5. Open land/mudflats situated between the forest boundaries and the sea; and
6. communities whose livelihoods are based on and dependent on mangrove areas.

Mangrove ecosystems provide a multitude of services: acting as habitats for diverse life forms; filtering pollutants from land and sea; trapping sediment; protecting coastlines from high waves, storms, and tsunamis; controlling beach abrasion; offering recreational spaces with beautiful natural scenery; and providing resources for fisheries and ecotourism. Additionally, they yield derivative products linked to local livelihoods and possess a high capacity for carbon storage (4-5 times that of mainland forests). Given the vital functions of mangrove ecosystems, their degradation and disappearance can have significant and long-term impacts on environmental sustainability and human well-being.

ASEAN boasts a coastline of 173,000 km, where an estimated 77 percent of the region's total population of 2.62 billion people reside in coastal areas (PEMSEA, 2015). In 2020, the total global mangrove area was approximately 14.8 million hectares (FAO, 2023), with Southeast Asia alone accounting for 5.1 million hectares, making it the largest region worldwide and covering around 33.5% of the world's total mangroves. Indonesia holds the largest mangrove area in both Southeast Asia and globally, with 3.36 million hectares. However, over the past two decades, there

has been a decline in mangrove cover across the ASEAN region, attributed to both natural phenomena (non-anthropogenic factors) and human activities (anthropogenic factors).

In the context of sustainable mangrove management within the ASEAN region, the ASEAN Strategy for Sustainable Mangrove Ecosystem Management (AMMS) was prepared by ASEAN Member States in 2024. This strategy aims to realize a shared vision and mission for sustainable mangrove ecosystem management across the ASEAN region for the period 2024-2029. To facilitate ASEAN countries in implementing this regional strategy, various technical guidelines have been developed, one of which focuses on preserving or protecting mangroves by designating them as Ramsar sites.

The preparation of this technical guideline is a detailed elaboration of Strategy 3 (*Strengthening governance as well as establishing and implementing appropriate policy related to mangrove ecosystem management in each AMS*), particularly Strategy 3, item number 6, which states: "*Preparing scheme and methods for incorporating program-related mangrove ecosystem management into the framework of the relevant global agenda, in particular Nationally Determined Contribution (NDC) of climate change convention, biodiversity convention, wetlands convention, and the UN Decade on Ecosystem Restoration*."

B. Goal and Objectives

The goal of this guideline is to achieve the sustainable protection and preservation of mangrove forest ecosystems in coastal areas of ASEAN countries. The purpose of this guideline is to facilitate parties in ASEAN countries in designating mangrove ecosystems as areas to be protected under the Ramsar Convention.

C. Scope

This guideline outlines the steps for determining and managing wetland areas, especially mangrove areas, as locations to be protected by the Ramsar Convention (Ramsar Sites).

D. Glossary

1. Mangrove

Vegetation that grows in coastal areas, characterized by distinctive morphology and a root system adapted to tidal zones with muddy or sandy-mud substrates.

2. Mangrove Ecosystem

A mangrove ecosystem is a natural system where life processes occur, reflecting the interrelationship between living organisms and their environment, as well as interactions among the organisms themselves. It is found in coastal areas, influenced by tidal movements, and is dominated by distinctive species of trees or shrubs that are capable of growing in saline or brackish water.

3. Mangrove Forest

A mangrove forest is a group of plant species that grow along tropical to subtropical coastlines, possessing special functions in environments containing salt and characterized by coastal landforms with anaerobic soil conditions.

4. Ramsar Convention

The Ramsar Convention is an agreement among member countries to commit to the conservation of the ecological processes of internationally important wetlands and to strive for the wise management of wetland areas within their territories based on the principles of sustainable use.

a) Ramsar Information Sheet (RIS)

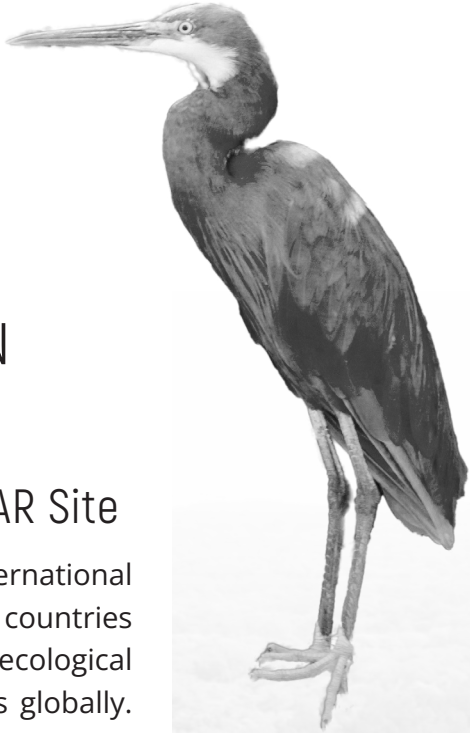
A standardized document used to describe and register a wetland into the List of Wetlands of International Importance under the Ramsar Convention on Wetlands.

b) Wetlands

As defined by the Convention, wetlands include a wide variety of habitats such as marshes, peatlands, floodplains, rivers and lakes, and coastal areas such as saltmarshes, mangroves, and seagrass beds, but also coral reefs and other marine areas no deeper than six metres at low tide, as well as human-made wetlands such as waste-water treatment ponds and reservoirs.

II.

RAMSAR CONVENTION



A. Background of RAMSAR Site

The Ramsar Convention is an international agreement among member countries committed to maintaining the ecological processes of important wetlands globally. Members also strive to manage wetlands within their territories wisely, based on principles of sustainable utilization. The Convention is named after the city of Ramsar in Iran, where it was formulated in 1971.

Unlike many other global environmental agreements, this Convention is not affiliated with the UN's multilateral agreements. While the UN and UNESCO act as Depositories of the Convention, it is not part of UNESCO. The Convention reports solely to its Conference of the Parties (COP). For management purposes, the Standing Committee, formed by the COP, selects the members of the Secretariat,

who carry out daily administrative tasks. Because it is appointed by the COP, the Secretariat is accountable to the COP, not the UN or UNESCO. The Convention's Secretariat is currently based in Gland, Switzerland.

Initially, the Ramsar Convention primarily focused on the conservation of waterbirds and migratory birds. However, over time, its scope evolved, and the emphasis shifted to broader wetland conservation. There was also a growing awareness of biodiversity integrity and the importance of wise use principles for wetlands. The Convention's main objective is to halt the encroachment upon and destruction of wetlands, which are fragile and sensitive ecosystems. Their very existence is highly dependent on how they are utilized and the efforts made towards their preservation.

The original text of the Ramsar Convention has been amended twice during its development: first by the Paris Protocol of 1982, and subsequently by the Regina Amendment of 1987.

The Paris Protocol was adopted at an extraordinary Conference of the Contracting Parties (COP) meeting held at UNESCO's Paris headquarters on December 3, 1982. This Protocol officially came into force in 1986. It governs the procedures for amending the Convention (Article 10 bis) and for ratifying the Convention text in multiple languages (Arabic, French, English, German, Russian, and Spanish).

The Regina Amendment (also known as The Second Amendment) was adopted during a regular meeting held in Regina, Canada, in 1987. Generally, this amendment did not alter the fundamental substance of the Convention. Instead, it primarily addressed operational issues, including the authority of the COP, the

establishment of the Standing Committee, budgeting, the formation of the Bureau, and the Ramsar Secretariat. The Regina Amendment took effect on May 1, 1994.

B. Definition of RAMSAR Site

The Ramsar Convention defines wetlands as 'areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six meters. This definition is precisely outlined in Article 1, paragraph (1) of the Ramsar Convention.

A wetland ecosystem is a transitional ecosystem situated between aquatic and terrestrial environments. Its existence is characterized by the dominance of a water regime and the presence of plants (hydrophytes) that possess strong adaptation capabilities to perpetually saturated (flooded) land conditions. Physically, it takes the form of areas as defined in Article 1 (Finlayson, C.M. et al., 1995).

A thorough understanding of wetland ecosystem characteristics is essential to ensure that their utilization remains mindful of their nature and character, thereby minimizing ecological impact. Wetland ecosystems can be grouped into freshwater ecosystems and estuarine (estuary) ecosystems. Freshwater ecosystems can include still water bodies such as swamps, ponds, and pools, as well as flowing water like rivers. Estuarine ecosystems are influenced by the ebb and flow of seawater, exemplified by brackish water swamps, mangrove forests, and lagoons.

Additionally, wetland ecosystems can be categorized into natural and artificial wetlands. These are further grouped into three main categories: marine and coastal wetlands, inland wetlands, and human-made wetlands (Scott D.A. et al., 1995).

Referring to this definition, mangrove ecosystems that grow and develop in coastal areas and are influenced by the ebb and flow of seawater are part of the wetland area that can be designated as an independent Ramsar Site or in conjunction with other ecosystems.

C. Basis For Determining RAMSAR Sites

The designation of a Ramsar Site is an integral part of The Ramsar Convention, an international agreement dedicated to the conservation and wise use of wetlands. This Convention aims to conserve and sustainably utilize wetlands, thereby preventing their loss. The background to the designation of a Ramsar Site includes:

1. International Agreement: The Ramsar Convention, signed in 1971 in Ramsar, Iran, is a treaty that governs national actions and international cooperation concerning wetland conservation.
2. Wetland Protection: The primary goal of designating a Ramsar Site is to protect wetlands of international importance, particularly those serving as habitats for waterbirds.
3. Sustainable Use: The Convention also emphasizes the wise and sustainable use of wetlands, ensuring their ecosystems are not damaged.
4. Global Efforts: The Ramsar Convention enjoys broad international support, including from ASEAN countries, which are committed to protecting and managing their wetlands.

5. Importance of Wetlands: Wetlands play a crucial role in maintaining environmental balance by, for instance, preserving water quality, mitigating flood risks, and serving as habitats for diverse plant and animal species.

Wetlands are areas with a high concentration of most species of birds, mammals, reptiles, amphibians, fish, and invertebrates. They also play a crucial role for various types of flora that depend on them. Some flora holds immense global value, such as rice, which is consumed by over half of the world's population and thrives in wetland environments.

Furthermore, wetlands are vital in supplying human needs for clean water. The growth of civilization and the global population has led to significantly increased clean water consumption. It is estimated that this consumption will rise even further as the world's population grows until 2025, potentially leading to high water stress conditions.

These areas also contribute to maintaining climate balance and fulfilling the economic needs of surrounding communities. Such needs can be met through the utilization of wetland- dependent flora and fauna (like fish, shrimp, and crabs) or through their tourism potential.

Unfortunately, the significant role of wetlands is often undervalued. The challenge of quantifying the economic value of wetland ecosystems in a publicly acceptable manner increasingly jeopardizes these areas. Land conversion, pollution, destruction, and over- exploitation of wetlands pose serious threats to their future sustainability and existence. Integrated efforts are therefore essential, not only within individual countries but also through cross-country cooperation, to halt the destruction of the

world's wetlands. International cooperation is particularly crucial for protecting wetlands that possess ecological characteristics spanning more than one nation.

D. Purpose of Determining A RAMSAR Site

The primary purpose of Ramsar Site designation is to protect wetland areas of significant ecological value worldwide. It aims to foster the conservation and wise use of wetlands through both national action and international cooperation.

Ramsar Site designation also seeks to conserve biodiversity, particularly waterbird habitats, and to ensure that wetlands function effectively as water sources, water purifiers, and coastal protectors.

The Ramsar Convention encourages the designation of wetlands that are representative, rare, or unique, or those that are important for conserving biological diversity. Wetlands meeting specific criteria may be added to the Ramsar List by the relevant national authority.

Ultimately, another key aim of designating a Ramsar Site is to contribute to sustainable development globally, while simultaneously raising awareness about the vital importance of wetlands for both human life and the environment.

E. Types and Classification of Wetlands In RAMSAR

Under the Ramsar Convention, wetland types have been defined to provide a very broad framework, aiding the rapid identification of the principal wetland habitats present at each Ramsar site. Wetland types are identified for each site on its relevant Ramsar Information Sheet.

The codes used to define wetland types for Ramsar sites are based on the Ramsar Classification System for Wetland Types, which was approved by Recommendation 4.7 and subsequently amended by Resolutions VI.5 and VII.11 of the Conference of the Parties. The following are the types of wetlands recognized by the Ramsar Convention:

1. Marine/Coastal Wetlands

a) Permanent shallow marine waters in most cases less than six meters deep at low tide; include sea bays and straits.

b) Marine subtidal aquatic beds: Including seagrass beds and tropical sea meadows.

c) Coral reefs.

d) Rocky marine shores: Including rocky offshore islands and sea cliffs.

e) Sand, shingle, or pebble shore: sincludes sand bars, spits and sandy islets; includes dune systems and humid dune slacks.

f) Estuarine waters: permanent water of estuaries and estuarine systems of deltas.

g) Intertidal mud, sand, or salt flats.

h) Intertidal marshes: Includes salt marshes, salt meadows, saltings, raised salt marshes; includes tidal brackish and freshwater marshes.

i) Intertidal forested wetlands: Including mangrove swamps, nipah swamps, and tidal freshwater swamp forests.

j) Coastal brackish/saline lagoons: Brackish to salty lagoons with at least one relatively narrow connection to the sea.

k) Coastal freshwater lagoons: brackish to saline lagoons with at least one relatively narrow connection to the sea.

l) Karst and other subterranean hydrological systems, marine/coastal

2. Inland Wetlands

1. Permanent inland deltas.
2. Permanent rivers/streams/creeks; include waterfalls.
3. Seasonal/intermittent/irregular rivers/streams/creeks.
4. Permanent freshwater lakes (over 8 ha); include large oxbow lakes.
5. Seasonal/intermittent freshwater lakes (over 8 ha); include floodplain lakes.
6. Permanent saline/brackish/alkaline lakes.
7. Seasonal/intermittent saline/brackish/alkaline lakes and flats.
8. Permanent saline/brackish/alkaline marshes/pools.
9. Seasonal/intermittent saline/brackish/alkaline marshes/pools.
10. Permanent freshwater marshes/pools; ponds (below 8 ha), marshes and swamps on inorganic soil, with emergent vegetation water-logged for at least most of the growing season.
11. Seasonal/intermittent freshwater marshes/pools on inorganic soils; include sloughs, potholes, seasonally flooded meadows, sedge marshes.
12. Non-forested peatlands; include shrubs or open bogs, swamps, fens.
13. Alpine wetlands; includes alpine meadows, temporary waters from snowmelt.
14. Tundra wetlands include tundra pools, temporary waters from snowmelt.
15. Shrub-dominated wetlands; shrub swamps, shrub-dominated freshwater marshes, shrub carr, alder thickets on inorganic soils.

16. Freshwater, tree-dominated wetlands; include freshwater swamp forests, seasonally flooded forests, wooded swamps on inorganic soils.
17. Forested peatlands; peat swamp forests.
18. Freshwater springs; oases.
19. Geothermal Wetlands.
20. Karst and other subterranean hydrological systems, inland.

Note: "floodplain" is a broad term used to refer to one or more wetland types, which may include examples from the R, Ss, Ts, W, Xf, Xp, or other wetland types. Some examples of floodplain wetlands are seasonally inundated grassland (including natural wet meadows), shrublands, woodlands and forests. Floodplain wetlands are not listed as a specific wetland type herein.

3. Human-made Wetlands

1. Aquaculture ponds; Examples: Fishponds or shrimp ponds.
2. Ponds; including agricultural ponds, livestock ponds, small tanks (generally under 8 ha).
3. Irrigated land; including irrigation channels and rice fields.
4. Seasonally flooded agricultural land; including wet grasslands or intensively managed or grazed fields.
5. Salt exploitation sites; salt pans, brine pits, etc.
6. Water storage areas; reservoirs/dams (generally over 8 ha).
7. Excavations; gravel/brick/clay pits; excavation pits, mine holes.
8. Wastewater treatment areas; agricultural waste processing, sedimentation ponds, oxidation ponds, etc.
9. Canals and drainage channels, ditches.

   Zk (c). Karst and other subterranean hydrological systems, human made.

F. Area and RAMSAR Site

By 2018, there were already 2,303 Ramsar sites, covering an area of 228 million hectares, registered by 169 member states across six Ramsar Regions: Africa, Asia, Europe, Latin America and the Caribbean, North America, and Oceania. By 2025, the number of Ramsar sites worldwide has increased to 2,532, with a total area exceeding 257 million hectares.

Within ASEAN, Indonesia boasts the largest Ramsar site area, totaling 1,375,426 hectares, distributed across 7 different sites. The Philippines ranks second with an area of 431,890 hectares. Thailand is among the top three, with its Ramsar sites covering 405,219 hectares. Myanmar follows in fourth place with a total Ramsar site area of 278,913 hectares. Malaysia holds the fifth position with a Ramsar site area of 134,182 hectares.

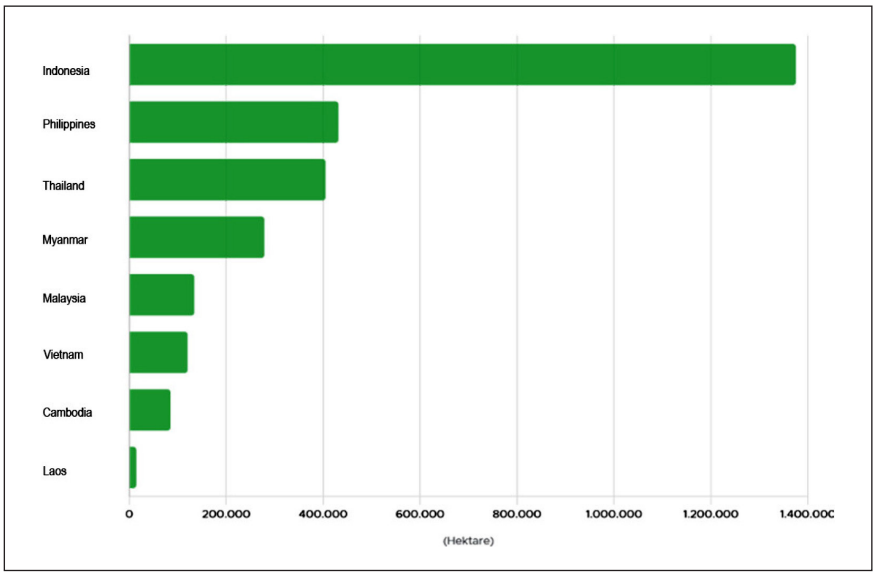


Figure 1. Area of Ramsar Sites in ASEAN Countries

Vietnam holds the sixth position with a Ramsar site area of 120,549 hectares. Cambodia follows in seventh place, with its Ramsar sites covering 85,235 hectares. Laos is eighth with a Ramsar site of 14,760 hectares. There are three countries in the ASEAN region that are not party to the Ramsar Convention or do not have Ramsar sites: Singapore, Timor Leste, and Brunei Darussalam.

G. Potential of Mangrove as A RAMSAR Site

The total area of mangroves globally 020 was approximately 14.8 million hectares (FAO, 2023). Southeast Asia accounts for 5.1 million hectares, making it the largest region in the world for mangroves, covering around 33.5% of the global total. Indonesia itself possesses the largest mangrove area in both Southeast Asia and the world, with 3.36 million hectares.

Box 1. Designation of Pulau Rambut Wildlife Sanctuary, Indonesia as a Ramsar Site

Pulau Rambut was designated as a Ramsar Site due to its rich biodiversity, particularly as a habitat for migratory birds, including those resting along the East Asian Flyway migration route. Pulau Rambut Wildlife Sanctuary also serves as an example of a natural mangrove ecosystem crucial for the protection of flora and fauna.

Pulau Rambut Wildlife Sanctuary was officially declared the 1987th RAMSAR SITE globally and the sixth in Indonesia, following Wasur National Park, Danau Sentarum National Park, Berbak National Park, Rawa Aopa Watumohai National Park, and Sembilang National Park. This designation took place on Friday, November 11, 2011.

Pulau Rambut is home to 73 bird species, including waterbirds that use the island as a resting place during migration. It is estimated that Pulau Rambut Wildlife Sanctuary is inhabited by over 20,000 individual waterbirds and migratory birds. Pulau Rambut is an example of a pristine ecosystem, vital for maintaining biodiversity and protecting endangered species.

Ramsar Sites that include mangrove ecosystems encompass various regions worldwide that have been designated as internationally important wetlands. Examples include Pichavaram in Tamil Nadu, India; Bhitarkanika Mangroves in Orissa, India; Del Carmen Mangrove Reserve in Siargao, Philippines; and sites in Indonesia, among others. These sites are crucial for protecting wetland functions and mangrove ecosystems, which provide numerous benefits, including coastal protection, habitat for diverse species, and carbon sequestration.

Examples of Ramsar Sites with Mangrove Ecosystems:

- **Pichavaram, India**: A small coastal village in Tamil Nadu, located between the Vellar and Coleroon estuaries, encompassing a protected mangrove forest area.
- **Bhitarkanika Mangroves, India**: One of the finest mangrove forests in India, providing a habitat for various species, including saltwater crocodiles and migratory birds.
- **Del Carmen Mangrove Reserve, Philippines**: A large mangrove area on Siargao Island, offering important habitat for various species and providing coastal protection.
- **Amazon Estuary and its Mangroves**: The mangrove area at the mouth of the Amazon River, also designated as a Ramsar Site.

- **Mangroves de Tsiribihina, Madagascar**: A mangrove area with diverse bird species, including rare and threatened ones, as well as habitat for various other species.
- **Guangdong Shenzhen Futian Mangrove Wetlands, China**: A mangrove area that is part of the Guangdong Futian Mangrove Nature Reserve.
- **Parc national des Mangroves, Democratic Republic of Congo**: A Ramsar site protecting mangrove forests in Congo.
- **Pulau Rambut Wildlife Sanctuary, Indonesia**: A site that protects waterbirds within a mangrove area.
- **Sembilang National Park, Indonesia**: This national park site includes a mangrove ecosystem covering an area of 94,622.05 hectares as one of its contributing ecosystems.
- **Rawa Aopa Watumohai National Park, Indonesia**: The extent of mangroves in this National Park is 6,173 hectares.

Locations to be protected by the Ramsar Convention (Ramsar Sites).

Box 2. Puerto Princesa Subterranean River National Park

Puerto Princesa Subterranean River National Park. 30/06/12; Palawan; 22,202 ha; 10º10_00_N 118º55_00_E. National Park, National Geological Site, ASEAN Heritage Park, UNESCO Biosphere Reserve and World Heritage Site. Located on the island of Palawan, the site is unique in the biogeographic region because it connects a range of important ecosystems from the mountain-to-the-sea, including a limestone karst landscape with a complex cave system, mangrove forests, lowland evergreen tropical rainforests, and freshwater

swamps. It is home to about 800 plant and 233 animal species, including the critically endangered Philippine cockatoo (*Cacatua haematuropygia*) and Hawksbill turtle (*Eretmochelys imbricate*), as well as the endangered green sea turtle (*Chelonia mydas*) and Nordmann's greenshank (*Tringa guttifer*). There are also some 15 endemic species of birds such as the Palawan peacock pheasant (*Polyplectron emphanum*) and the Tabon scrub fowl (*Megapodius freycinet cumingii*). One of the unique features of the park is an 8.2 km long section of the Cabayugan River that flows underground within large formations of stalactites and stalagmites. The river provides water to local communities for domestic and agricultural uses, before flowing towards the underground river. The site is a major ecotourism destination, and community- based sustainable ecotourism has been initiated to involve the local communities in Park management as well as to generate income.

Ramsar Site no. 2084. Most recent RIS information: 2012

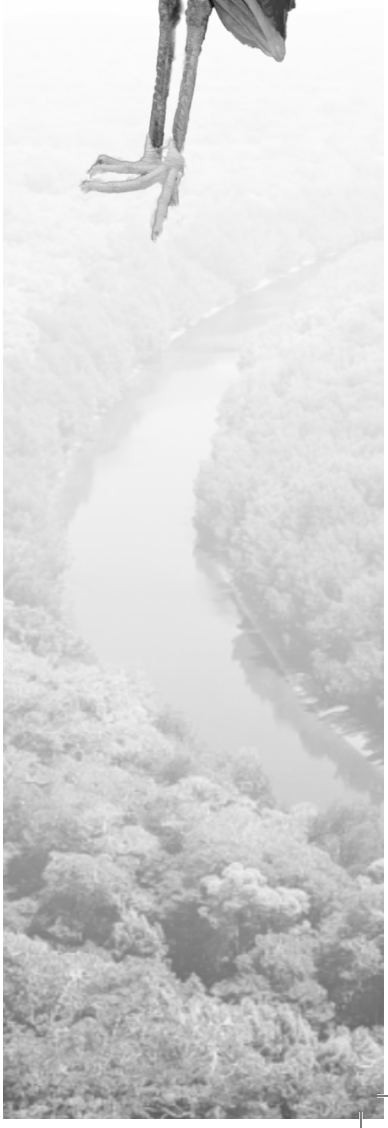
III.

DETERMINATION OF RAMSAR SITES

A. Determination Criteria

Mangrove ecosystems can be designated as Ramsar Sites if they meet specific criteria, such as possessing significant ecological value, high biodiversity, and serving as important habitats for various species, including waterbirds.

The criteria used to identify internationally important wetlands are divided into nine categories. These criteria can be found in the 'Strategic Framework and guidelines for the future development of the List of Wetlands of International Importance,' which is available in the 14th handbook of the *Ramsar Handbooks for the Wise Use of Wetlands*. The nine criteria cover the uniqueness of the wetland type, its species community and ecology, as well as waterbirds, fish, and other taxa. The agreed wetland criteria are as follows:



1. Group A: Representativeness, Rarity, or Uniqueness

Criterion 1: The wetland is a particularly good example of a natural or near-natural wetland type found within the appropriate biogeographic region or other nearby regions.

2. Group B: Biodiversity Conservation

a) Criteria based on species and community ecology.

10. Criterion 2: The wetland supports vulnerable, endangered, or critically endangered species or threatened ecological communities.
11. Criterion 3: The wetland supports populations of plant and/ or animal species important for maintaining the biodiversity of a particular biogeographic region.
12. Criterion 4: The wetland supports plant and/or animal species at a critical stage in their life cycle or provides refuge during adverse conditions.

b) Criteria based on the presence of waterbirds.

1. Criterion 5: The wetland regularly supports 20,000 or more waterbirds.
2. Criterion 6: The wetland regularly supports 1% of the individuals in a population of one species or subspecies of waterbird.

c) Criteria based on the presence of fish

1. Criterion 7: The wetland supports a significant proportion of indigenous fish subspecies, species or families, their life-history stages, interactions and/or populations that are representative of wetland benefits and/or contribute to global biological diversity.

2. Criterion 8: The wetland is an important source of food for fish, a spawning ground, a nursery, and/or a migration path on which fish depend, whether within the wetland or in other related areas.

d) Criteria based on the presence of other taxa

1. Criterion 9: The wetland regularly supports 1% of the individuals in a population of one species or subspecies of other taxa (besides fish and waterbirds) that are dependent on the wetland for their survival, whether at the species or subspecies level.

3. RAMSAR Site Proposal Process

The process of designating a Ramsar site begins with identifying a wetland area that meets the criteria, followed by a determination process involving local parties and obtaining approval from the Ramsar authority. The wetland must satisfy at least one of the nine criteria to become a Ramsar site. Once designated, the site is added to the Convention's List of Wetlands of International Importance.

Here are the more detailed steps:

1. **Area Identification:** Potential wetlands to become a Ramsar site are identified based on criteria established by the Ramsar Convention on Wetlands.
2. **Criteria Evaluation:** The area is assessed against the nine existing criteria, such as biodiversity, representativeness, and ecological value. Guidelines for assessing biodiversity in mangrove ecosystems can be found in the Physical, Biological, and Socioeconomic Survey Guidelines for Mangrove Ecosystems.

3. **Local Approval:** The designation process involves the participation and support of local communities, as they are the ones closest to the wetlands.
4. **Authority Agreement:** The Authority, representing the countries that are parties to the Ramsar Convention, provides agreement on the designation.
5. **Registration:** After receiving consent, the site will be registered in the List of Wetlands of International Importance of the Ramsar Convention.
6. **Management:** Wetlands that have become Ramsar sites need to be managed properly to en- sure their sustainability. The designation of Ramsar sites aims to protect and preserve wet- lands that have high ecological value. These wetlands play a crucial role in various aspects, including biodiversity conservation, water regulation, and carbon sequestration.

B. RAMSAR Documents

The Ramsar Convention encourages the development of three key documents for each Ramsar site, namely:

1. Ramsar Information Sheets (RIS)

Contracting Parties to the Ramsar Convention are required to provide a Ramsar Information Sheet (RIS) for all sites designated as wetlands of international importance under the Convention. The RIS must be submitted to the Ramsar Secretariat at the time a site is nominated to the List of Wetlands of International Importance. Furthermore, the parties involved in the Ramsar Convention are committed to providing updated RIS information for all their Ramsar sites at six-year intervals, or whenever a significant change in the site's ecological character occurs.

The RIS provides crucial data about every wetland designated as an internationally important wetland site. This allows for analysis of wetlands registered worldwide under Ramsar, offers baseline data for measuring changes in the ecological character of wetlands listed under the Ramsar Convention, and provides material for publications that inform the public about Ramsar sites.

More detailed information regarding RIS, and information about the format, is available on the Ramsar Convention Website under 'Explanatory Note and Guidelines for completing the Information Sheet on Ramsar Wetlands'.

2. Ecological Character Descriptions (ECDs)

An Ecological Character Description describes the ecological characteristics of a wetland at the time it is registered as a Wetland of International Importance. Ecological Character Descriptions for wetlands of Ramsar sites are typically used to:

1. Assist in the implementation of each country's obligations in the management of Ramsar sites under the Ramsar Convention.
2. Describe and maintain the ecological character of designated Ramsar wetlands in each country.
3. Serve as a basis for formulating and implementing planning, management, and monitoring that supports: * wetland conservation, and * wise and sustainable use of wetlands for the benefit of humanity in a manner consistent with maintaining the natural character of the ecosystem.
4. Assist in fulfilling the obligation of the State hosting the Ramsar site under the Ramsar Convention to arrange for notification as early as possible if the ecological character

of any wetland in its territory has changed, is changing, or is likely to change as a result of technological development, pollution, or other human intervention.

5. Complement the description of ecological character contained in the Ramsar Information Sheet (RIS) submitted under the Ramsar Convention for each listed wetland and, collectively, form the official record of the site's ecological character.
6. Provide material to inform the general public interested in declared Ramsar wetlands, helping them to understand and appreciate these sites.

C. Management Plans

Management Plans are utilized to formulate and implement strategies that promote the wise use and conservation of wetlands. Ramsar site management plans must adhere to international provisions and regulations within each respective country.

The summarized framework for data and information requirements pertaining to primary inventory, ecological characteristics data collection, RIS completion, Ramsar site designation, and reporting is illustrated in Figure 2.

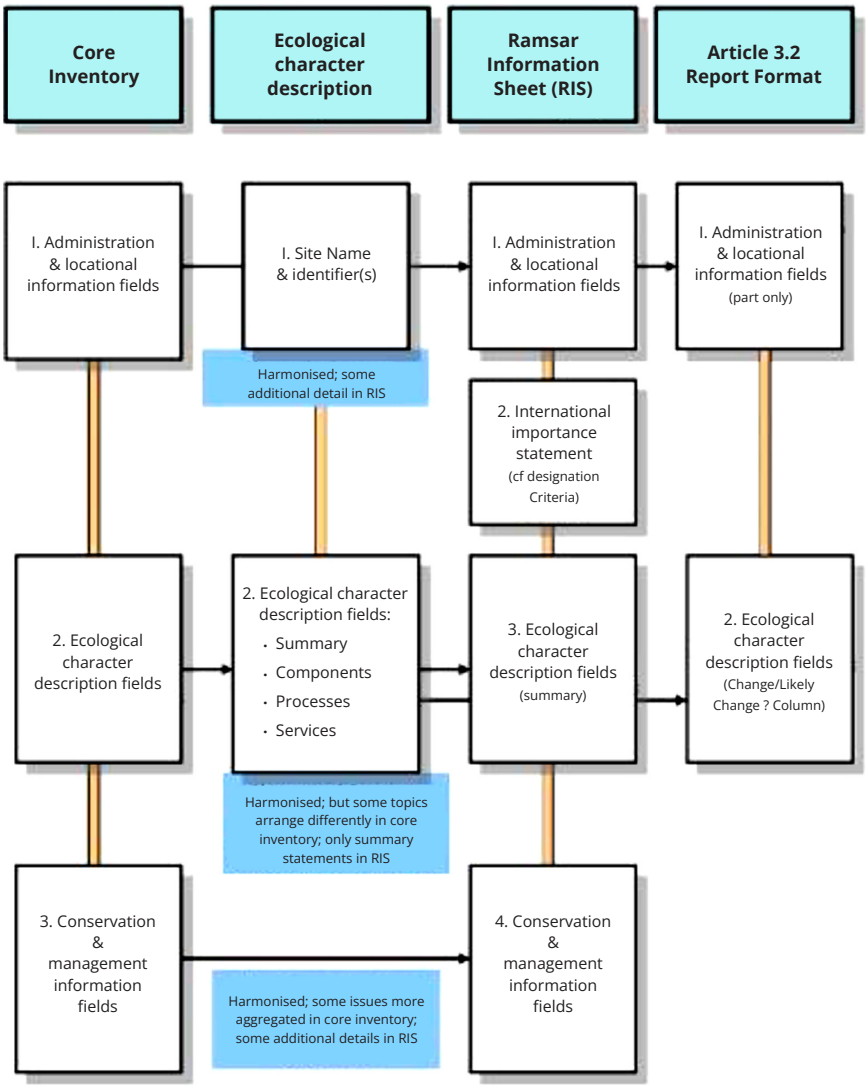
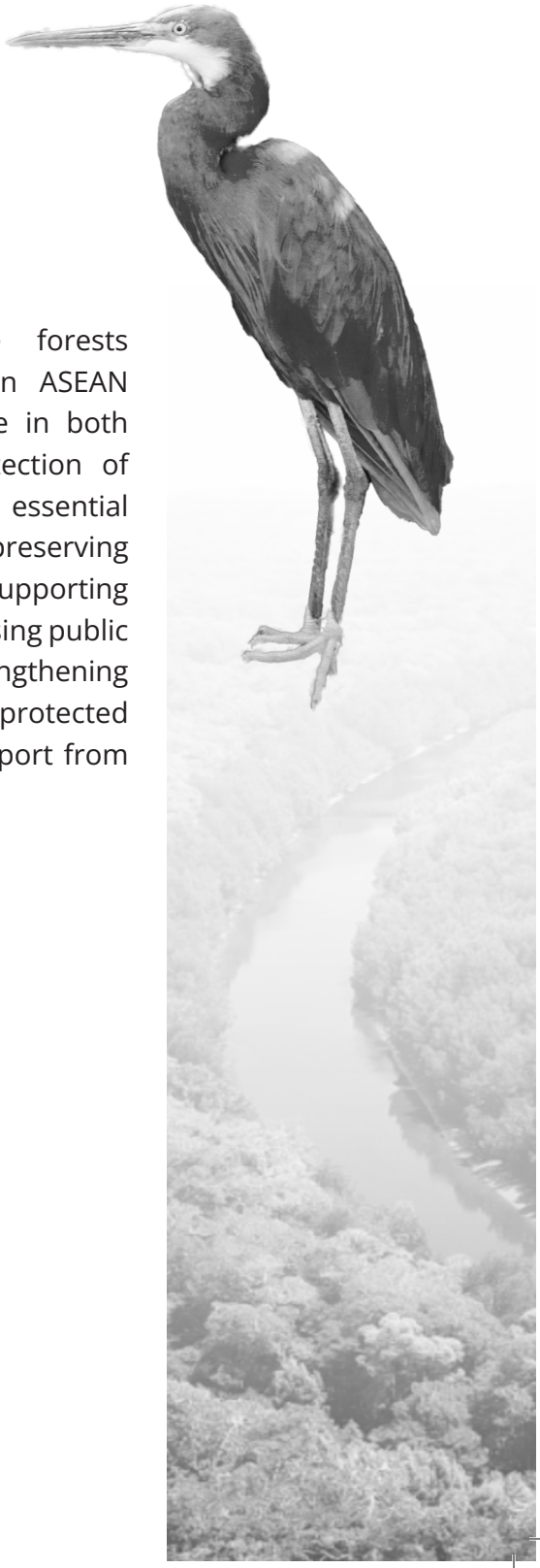


Figure 2. Summary of the Framework for Data and Information Needs for Inventory, Ecological Character Description, Ramsar Site Designation, and Article 3.2 Reporting

IV.

CLOSING

The condition of mangrove forests around the world, including in ASEAN countries, continues to degrade in both quantity and quality. The protection of mangrove ecosystems, as an essential part of wetlands, is crucial for preserving and enhancing biodiversity, supporting sustainable development, increasing public awareness and participation, strengthening the world's largest network of protected areas, and securing funding support from various stakeholders.



REFERENCES

Aksornkoae. (1993). Ecology and Management of Mangroves. IUCN Bangkok, Thailands. 176 pp.

Anggara, AS. Aspek Hukum Pelestarian Lahan Basah pada Situs Ramsar di Indonesia (Studi terhadap Implementasi Konvensi Ramsar 1971 di Taman Nasional Tanjung Puting). *MIMBAR HUKUM* 30 (2), Juni 2018, Halaman 246–261.

Finlayson, CM *et al*. (1995). Classification and Inventory of The World's Wetlands, Springer, hlm.41.

Macnae W. (1968). A General Account of the Fauna and Flora of Mangrove Swamps and Forest in the Indo West-Pacific Region. *Adv. Mar. Biology.* Publikasi pada http://www.ramsar.org.

Ramsar Convention Secretariat. (2016). An Introduction to The Convention on Wetlands, Ramsar Convention Secretariat, Gland, Switzerland.

Richards, Daniel R., Friess, Daniel A., et al. (2015). Rates and drivers of mangrove deforestation in Southeast Asia, 2000–2012. Department of Geography, Nation- al University of Singapore. PNAS.

Scott DA, et al. (1995). Classification and Inventory of Wetlands: A Global Overview, *Vegetation*, 118(1/2), hlm.10.

Snedaker. (1978). Mangrove; Their Values and Perpetuation. *National Resources.* 14:6–13.

Ulqodry, TJ, AE Aprianto, A Agussalim, R Aryawati, dan A Absori. (2021). Analisis Tutupan Mangrove Taman Nasional Berbak–Sembilang melalui Citra Landsat-8 dan Pemantauan Leaf Area Index (LAI). *Jurnal Kelautan Tropis* 24(3): 393–401.